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Game, guide or website for financial education improvement: Evidence from an experiment in Colombian schools

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ABSTRACT

This paper evaluates the impact of a financial education program consisting of three different learning tools, for children (6–18) in Bogota, Colombia, in 2018. This randomized experiment offers the opportunity to examine the impact of those three learning tools (Game, Guide, Website) on students' financial knowledge and literacy on a sample of 1337 students from four Colombian schools, and four academic grades. We did not find a significant general treatment effect; however, we did find an improvement in treated public school students. Regarding the efficacy of the tools, we found several heterogeneous effects among the types of schools and grades. The type of school (public/private) is relevant in terms of knowledge improvement and still represents a pre-existing condition that leads to differences in children's performance, but implementing a program can help mitigate these differences. We found that differentiated instruction in the learning process can be effective even if significant pre-existing differences still prevail. This study contributes to creating a framework for the current situation of Colombian children's financial literacy and serves as insight for future programs.

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1. Introduction

Financial literacy has recently become more relevant both academically and for policymakers (Lusardi et al., 2010) because it has an impact on short- and long-term budgeting decisions (Kim et al., 2019). There is a close and important relationship between financial literacy and financial inclusion. Research has identified that financial literacy is one of the factors that can improve financial inclusion, which in turn, leads to people and countries' greater economic and financial wellbeing (Beck et al., 2007; García et al., 2013). There is a direct relationship between the country's and its people's financial literacy, financial inclusion and economic wellbeing. Atkinson and Messy (2013) point out that 2.3 billion working age adults are financially excluded. Several factors, from geographical barriers to high costs of financial products or services, lead to this situation. However, one of the main reasons is the lack of financial literacy. Regarding the relationship between inclusion and economic wellbeing, Beck et al. (2007) found a positive relationship between financial development and greater income equality among the population, especially in its poorest quintile of the population. García et al. (2013) analyze the situation in Latin America and the Caribbean and state that "a 10% increase in access to financial services generates a 0.6 point reduction in the Gini inequality coefficient, while an increase 10% in private credit reduces poverty by about 3%".

The OECD defines financial education as: "The process by which consumers or investors improve their understanding of financial products, concepts and risks, and through information, instruction and/or objective advice, develop the skills and confidence to be more aware of financial risks and opportunities, make informed decisions, know where to go for help and take any effective action to improve their economic wellbeing" (OECD INFE, 2011). This definition implies that, financial education allows people to enjoy greater financial wellbeing. There have been several initiatives to improve financial knowledge, one of which is to teach young school children (Amagir et al., 2018; Bruhn et al., 2016). The OECD INFE (OECD INFE, 2012) insists that financial education in children is essential as they are the adults of tomorrow, and that children and the youngsters are potentially good at disseminating new habits in financial education among the rest of the population. Friedline et al. (2015) identify a number of reasons for starting financial education from an early age. There is a growing body of literature that focuses on the implementation of these financial education programs (Iterbeke et al., 2020; Kaiser and Menkhoff, 2020; Kaiser et al., 2020). Thus, further study of

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the implementation of programs and their efficacy is essential in order to allocate resources properly.

The OECD Program for International Student Assessment (PISA) evaluates knowledge and skills in young people worldwide to uncover the disparities existing among countries. In particular, this assessment found that Colombia's level is low compared to other countries, highlighting the need to tackle a financial education curriculum as a need in schools. Few studies and program implementations have been conducted on financial education for children and youth in Colombia. This study helps to create a framework for the situation of Colombian children, and serves as insight for education programs. We use a randomized experiment to examine the impact of three learning tools (Game, Guide, Website) on students' financial knowledge and literacy on a sample of 1337 students from four Colombian schools, and four academic grades.

For this research, we applied surveys at four schools in Colombia's capital, Bogota, to work with children and youth in second, fourth, seventh and tenth grade,[1] for a total of 1337 participants. We designed four different surveys, according to grade and age, where the questions measured the knowledge of the children and adolescents. In addition, we implemented an educational program over twelve weeks, which consisted of three tools: a Game, a Website, and a Guide. We compared pretest and posttest data from the students' to record any changes in their level of knowledge. We also analyzed which tool was the most effective by comparing the data to those of the control group. In this paper, we explore three main questions. First, does the finance education program have an impact on the level of knowledge of Colombian school children? Second, which of the three tools is more efficient in terms of improving financial knowledge? And third, are there any heterogeneous effects on the students regarding the type of school or grade?

We did not find a significant overall treatment effect; however, we did find an improvement in treated public school students. Regarding the efficacy of the tools we found several heterogeneous effects among the types of schools and grades. In some cases, the more traditional tools seem to work better; for example, the Guide for 7th and 10th grades in the public school. While the more interactive tools seem to work better for 4th grade. The type of school (public/private) is decisive in terms of knowledge improvement and still represents a pre-existing condition that leads to differences in children's performance. We found that differentiated instruction on the learning process can be effective even if significant pre-existing differences still prevail. This study contributes to creating a framework for the current situation of Colombian children's financial literacy and serves as insight for future programs. The results are robust across different variable measurements.

The remainder of this paper proceeds as follows. In Section 2, we introduce the literature and context. In Section 3, we present the method used for the experiment, the design, the variables, and the behavioral predictions. In Section 4, we present the descriptive and empirical results. Finally, we discuss the conclusions in Section 5.

2. Literature and context

2.1. Financial education in children

Based on international experience, financial inclusion is crucial for individuals, but sound support in financial education is even more important (Raccanello and Guzmán, 2014). Research on financial education has involved all types of people, from school children, up to older adults. However, the OECD INFE shows the benefits of targeting children and younger people (OECD INFE, 2012). Evidence shows that financial education school programs can improve children and adolescents' financial knowledge (Amagir et al., 2018; Bruhn et al., 2016; Kaiser and Menkhoff, 2020; Sherraden et al., 2007). Van Campenhout (2015) compiles some results showing that acquiring knowledge and skills in financial education at a young age leads to greater future wealth, a better pension plan, and better debt management. Moreover, a positive exposure of students to financial education can also result in a future positive attitude towards personal finance and savings (Batty et al., 2015).

Even with variations in size, design and coverage of financial education programs, implementing programs for young people is a promising strategy, and an interesting method is to implement financial education programs for schoolchildren. Amagir et al. (2018), Kaiser and Menkhoff (2020) because this population is at a great learning age and the implementation can reduce spillover effects (Becchetti and Pisani, 2012; Berry et al., 2018; Bruhn et al., 2016; Frisancho, 2018; Iterbeke et al., 2020; Lührmann et al., 2015). A wide range of tools, designs, sizes and coverage of these studies shows that, on average, knowledge improves after being exposed to treatment, especially for this younger sample (Kaiser and Menkhoff, 2017, 2020), which contributes to supporting the importance of developing new programs and curricula. The success of the intervention depends not only on the teaching moment, but also on the intensity of the program (Kaiser and Menkhoff, 2017). Studies such as those conducted by Walstad et al. (2010) measure the level of financial literacy in high school students, before and after implementing a financial curriculum for youth. They found that knowledge increased in every grade in which it was used and across student characteristics. Along the same lines, Lührmann et al. (2015) evaluate the impact of a short program on financial education on young people in Germany. They show that the program increases their interest in financial matters as well as their financial knowledge. On the other hand, Iterbeke et al. (2020) find that a differentiated instruction approach on financial education is not effective in a school-based intervention because the initial heterogeneity in sociodemographic, family and peer characteristics seems to prevail.

It is worth mentioning that most of the tools used by financial education programs in different countries repeat various formats. Games, workshops, educational portals, and widely-covered websites are the tools most frequently used in financial education (Barrera Rodríguez and Rodríguez García, 2017; Denegri Coria and Sepúlveda Aravena, 2014; Guillén and Ayra, 2016). The workshops and booklets are designed primarily for segments of the population for which periodic interventions are required: generally, school-age children and youngsters, rural population, and people with some degree of disability. Over the years, the number of initiatives designed to teach financial issues through games on web portals has increased significantly, because a more interactive learning or more active learning has a greater success rate among children (Amagir et al., 2018; Iterbeke et al., 2020; Kaiser and Menkhoff, 2020). Even considering the differentiated effects among learning methods is useful to understand which one is the most effective (Iterbeke et al., 2020; Linden and MacLeod, 2008).

For Latin American in particularly in Peru, Vasquez Ruiz (2017) used a difference in differences impact evaluation program in primary school students, finding that knowledge and financial abilities had a positive and significant effect, but only in one of the grades in which the program was implemented. In Chile, a

[1] In Colombian schools, academic years (grades) go from one to eleven, grade one is the first of primary school, and eleven is the last of secondary school. Children begin school at the age of five and end at eighteen, on average.

program for students that used an economic literacy test found a positive and significant increase in financial literacy. It also found a difference between gender and the socioeconomic level (Denegri Coria and Sepúlveda Aravena, 2014). In Colombia, García Bohórquez (2012) analyzed the short-term impact of an economic and financial education program on 1518 high school students. She found that the program manages to increase knowledge, yet not financial literacy skills. But there is no concrete evidence on which tool would work best for students in private and public schools in Colombia (García et al., 2013; OECD INFE, 2012; Totenhagen et al., 2015).

2.2. Colombian context and educational system

Latin America generally obtains low PISA results compared to other countries (Gamboa Nino, 2012), some of which have managed to improve thanks to an effort to improve the education of children and youngsters at school. Colombia's results in the PISA tests have been low throughout the different measurements. There are deficiencies in the different areas, and this is directly related to the Colombian educational system (OECD, 2016).

In Colombia, education is a citizen's right and is provided for free through the public educational system (OECD, 2016). The Ministry of National Education (Colombia) sets the general policies, but the allocation of economic resources and the specific definition of the curricula rests with the Secretariats of Education of each municipality (Martínez Barrera, 2012; OECD, 2016). This does not allow for have national contents and a general standards.

Education in Colombia is divided into 12 academic grades. The grading system ranges from 0 to 5, with 3 being the passing grade. Schools can be public or private and the choice depends directly on the parents socioeconomic level (Orjuela, 2012). In Colombia, private schools serve only the 20% of the students (Martínez Barrera, 2012). The resources of public schools are obtained from the nation's budget, which, although it has increased in recent years, limits physical, technological and human resources. Private schools obtain income from tuition and its amount its defined by the school according to the socioeconomic level of the families that attend. An important difference between public and private schools is their school hours. Most public schools in Colombia have a double shift (morning and afternoon) with the purpose of covering a larger number of students, but reducing the time of contact with teachers. Private schools work with a single day shift and students spends more time at school (Martínez Barrera, 2012).

The results of the mandatory tests that the State has set for senior students show a significant difference between public and private school students, in terms of performance in math, reading, and other variables (Gamboa Nino, 2012; Castro Aristizabal et al., 2016). The disparities among the types of schools occur mainly due to the infrastructure, the teachers' effectiveness, the administration of the school system (Celis et al., 2012), the family's socioeconomic level (Orjuela, 2012), as well as the parents' educational level (Gamboa Nino, 2012).

2.3. The financial education program development

The Financial Education program developed in this research was carried out during 2018 in schools in Bogota (Colombia), but was in the making since 2011.[2] This is an original program, based on references from other programs and global surveys.[3] The contents of the program[4] and the selection of age ranges[5] were defined in the first instance in a work carried out by the authors in 2011.

Likewise, the surveys were constructed in an iterative process, considering the previous surveys, questions and programs. The tools were developed in sequential stages, starting with the Guides, in 2013 and the Website and Games, in 2017. Over time, some minor adjustments were made (survey questions, content, and ages), motivated by tests conducted in various schools. In particular, the Ministry of National Education (Colombia) reviewed the content and pedagogical aspects of the Guides in 2014. The comments made allowed us to confirm that the defined contents were suitable for the ages and capacities of children and youths in Colombian schools. The other two tools developed have the same contents as the Guides, which guarantees consistency between all of them.

3. Method

3.1. Research design and program

3.1.1. Program description

The Financial Education Program was created as an interactive program for children (6–18) to increase their level of financial literacy, and to be implemented in different schools and grades in Bogota, Colombia. The program consists on 3 interactive tools for children: a guide, a website and a board game.

All three tools cover the same topics, and introduces students to basic concepts such as saving, debt, taxes, decision making and banking. They all cover the same topics but at a level that is appropriate for each age group. Although previous studies worked with age groups (5–7, 8–11, 12–14, 15–18), we decided to choose grades that contained these age groups to ease implementation and avoid spillover effects. We chose to begin with 2nd grade because they already knew how to read, and ideally leaving three-grade gap between each group (grades 2, 4, 7 and 10). We made sure that the concepts of financial literacy covered in each grade tool and survey were consistent with the different ages.[6]

The first tool consists of a board Game with financial goals cards and money cards with questions at the back, where the highest value corresponds to a higher level of difficulty. Children choose one financial goal of a certain value. The student who answers the question correctly wins the money card, and is able to save up that money; the player that saves enough money to buy the financial goal, wins. There are two types of games, one in which the money cards are coins (for 2nd and 4th grades) and another in which money cards are bills (for 7th and 10th grades).

The second tool is a Website with twelve learning segments, each designed to be completed in 15 min and to be implemented during the computer class.

The third and final tool is a homework Guide with 10 sections, each of which is designed to be completed in 20 min or less. The guide was designed for use at home.

The main purpose of the different tools is to analyze whether any of the tools actually work to increase the students' knowledge of personal finance, and to define which tool works the best for this purpose.

2 See Appendix B Table B.14.
3 See Appendix B Table B.15.
4 See Appendix B Table B.16.
5 See Appendix B Table B.17.
6 Appendix C shows the content for each age/grade.

Table 1
Study design randomization.

	2nd Grade	4th Grade	7th Grade	10th Grade
Classroom 1	Guide 1	Guide 2	Guide 3	Guide 4
Classroom 2	Website 1	Website 2	Website 3	Website 4
Classroom 3	Game 1	Game 1	Game 2	Game 2
Classroom 4	Control	Control	Control	Control

3.1.2. Research design

We use a randomized control trial with three treatment arms, one for each tool and one control group, to measure the effects of the financial education program and the impact of each tool on financial knowledge. We assigned the tools, at school-grade-classroom level, to avoid a spillover effect between students. Table 1 shows an example of the randomization for one school. Table 2 shows each treatment and the corresponding tool level for each grade.

3.2. Schools sample selection

In 2017, Bogota had 729 public schools and 372 private schools, for a total of 1101 schools located in 19 localities. We looked for schools in different localities in Bogota (see the map D.2 in Appendix D), that had primary and high schools, with boys and girls, and grades that where large enough to include at least four classes in each grade. We also tried to have a representation of all socioeconomic levels of income, both in public and private schools. However, this sample is not intended to be representative of all schools in Bogota or Colombia.

While we preselected schools that met the requirements, and contacted public schools with the help of the Secretariat of Education of Bogota, while contacted the private ones directly. We explained both the purpose and the process of the investigation to the schools. Table 3 shows the different stages of our sample.

We were initially able to conduct the research in five schools in Bogota, Colombia, two public schools[7] and three private schools[8] Row 1 of Table 3 shows the intended to treat schools sample, with 2499 students, which we used for our initial randomization of the treatments.

During program implementation, one of the public schools decided to no longer participate, leaving us with four schools, one public[9] and three private,[10] as Row 2 of Table 3 shows. We lost 22.5% of our initial sample.[11]

3.3. Parents' consent

For this research we required three authorizations: one from the Ethics Committee of Universidad de los Andes,[12] one from the children's parents, authorizing their son/daughter's participation, and finally, the students' consent agreeing to join the project.

In the parents' consent form, in addition to the authorization, they also provided information such as age, educational level, type of current employment and their particular home situation, the social stratum to which they belong to, and the household size. We have no demographic information for parents who did not give their approval for participation, and therefore, we have no way of knowing whether they are systematically different from the parents who gave their consent, in terms of demographic characteristics.

Row 3 of Table 3 shows the intended kids' sample after parents gave consent to participate. We lost 21.5% of students, whose parents did not agreed to their participation. Since we do not have sociodemographic information to compare and analyze this attrition, Table A.13 in Appendix A shows the comparison between the full sample and the sample after consent, in terms of the variables we do have, which are: treatment assigned, grade and type of school. We can see that the consent rate for the 2nd grade is lower, but this is the only significant difference.

3.4. Implementation, attrition and randomization

3.4.1. Survey participation and attrition

We implemented the program over six months at public and private schools. The students in the sample were aged between 6 and 18 years of age. We applied a survey before implementing the program to determine the baseline level of knowledge, and tested the children again, following the program, to measure their level of improvement. Row 4 of Table 3 shows the baseline sample of children. After the parents' consent, we lost 10 students who did not answer the initial survey, however, this number is small.

Additionally, as with most pretest and posttest evaluations, we lost some information on those students that did not take the second test. The follow-up attrition rate was 11.5% for the whole sample. We have 91% of students in the follow-up survey assigned to the control group, 88% in the game treatment, 88% in the guide treatment, and 86% in the web treatment, which seems as a balanced rate for treatments.

Furthermore, we estimated a regression[13] of the probability of continuing in the sample on an indicator of whether the student was initially assigned to a specific tool, with classroom fixed effects. In addition, we estimated a similar regression, which also includes baseline characteristics. The results show no relation between remaining in the sample and the tool assignment in either specification. Moreover, baseline characteristics are not significantly correlated-neither individually nor jointly-with the likelihood of continuing in the sample. Thus, selection into the sample thus does not appear to be a problem.

3.4.2. Baseline summary statistics and randomization

Table 4 shows the summary statistics of the students and their parents characteristics for each of the three treatments and the control group. Overall, there is balance across treatments and control groups in variables such as participant's age, gender, father's age, education and having a savings account, as well as in stratum and household size, suggesting that the randomization is correct and supports internal validity.

3.4.3. Implementation

Table 5 shows the timeline of our implementation. A baseline survey was collected in March 2018 for all students and schools. In the first survey, students answered an initial test to see the baseline knowledge, in terms of financial literacy, as well as a survey with some demographic questions, such as age and gender. The financial education program was implemented thereafter.

The guide had to be completed within twelve weeks. Since the guide consisted of 12 sections, the students were supposed to cover one section per week, at home, which adds up to 3 hs

[7] Nueva Colombia (low income) and Marsella (medium income).

[8] Gimnasio Los Sauces (low income), Gimnasio Monseñor Manuel María Camargo (GMMMC) (medium income) and Anglo Americano (high income).

[9] Nueva Colombia.

[10] Gimnasio Los Sauces, Gimnasio Monseñor Manuel María Camargo (GMMMC) and Anglo Americano.

[11] Table A.12 in Appendix A shows the descriptive statistics of the data we have available, comparing the complete sample of 5 schools; not including Marsella School, reveals that losing the public school only modifies the rate of public schools in the sample, but that the control, treatments and percentage among grades remain the same.

[12] We were granted authorization from the Ethics Committee.

[13] We do not present these results, but they are available upon request.

Table 2
Program design: Tools and Grades.

Guide	Web	Game	Control
			
GuideL 1 for 2nd Grade Guide 2 for 4th Grade Guide 3 for 7th Grade Guide 4 for 10th Grade	Website 1 for 2nd Grade Website 2 for 4th Grade Website 3 for 7th Grade Website 4 for 10th Grade	Game 1 for 2nd & 4th Grades Game 2 for 7th&10th Grades	2nd Grade 4th Grade 7th Grade 10th Grade

Table 3
Sample stages.

1	Intended to Treat	Public		Private		Total
	Schools Sample	(n = 1122)		(n = 1377)		(n = 2499)
		Nueva Colombia (n = 559)	Marsella (n = 563)	Los Sauces (n = 582)	GMMMC (n = 406)	Anglo Americano (n = 389)
2	Intended to Treat	Public		Private		Total
	Kids Sample	(n = 559)		(n = 1377)		(n = 1936)
		Nueva Colombia (n = 559)		Los Sauces (n = 582)	GMMMC (n = 406)	Anglo Americano (n = 389)
3	Intended to Treat	Public		Private		Total
	Kids Sample	(n = 420)		(n = 1099)		(n = 1519)
	After Consent	Nueva Colombia (n = 420)		Los Sauces (n = 444)	GMMMC (n = 326)	Anglo Americano (n = 329)
4	Baseline Survey	Public		Private		Total
	Sample	(n = 416)		(n = 1093)		(n = 1509)
		Nueva Colombia (n = 416)		Los Sauces (n = 443)	GMMMC (n = 325)	Anglo Americano (n = 325)
5	Follow-up Survey	Public		Private		Total
	Sample	(n = 334)		(n = 1003)		(n = 1337)
		Nueva Colombia (n = 334)		Los Sauces (n = 427)	GMMMC (n = 312)	Anglo Americano (n = 264)

0. n corresponds to the number of students.
1. Full expected sample of schools, used for the randomization of the treatment. Marsella school decided to withdraw from the program, which translates into a sample loss of 22.5%. Table A.12 in Appendix A shows the descriptive statistics of the data we have available, comparing the complete sample of 5 schools; not including Marsella school, reveals that losing the public school only modifies the rate of public schools in the sample, but the control, treatments and percentage across grades remain the same.
2. After Marsella's withdrawal. The full sample of kids expected to participate in the project, assuming all parents give consent to participate.
3. The sample, after parents' consent, had 1519 students. The non-consent rate was 21.5%.
4. After parents' consent, we did not obtain the baseline survey for 10 students, leaving a sample of 1509 students.
5. Full data with information for baseline and follow-up survey only for students with parents' consent. Follow-up attrition rate was 11.4%.

in total for the treatment. Upon completing the guide, they had to hand in a deliverable at school for the teachers to verify that they have done so. Unfortunately, we have no information on the percentage of students that actually handed in this deliverable; therefore, we do not know the exact degree of compliance with the treatment. Additionally, we did not train parents in the specific topics of the guide, thus, the implementation and the work at home might vary across households.

The website was intended to be used during the computer class, which teachers helped us to implement it. The Website was self-contained and had all the required explanations for the students to complete. Students used the Website for approximately 15 min, once a week throughout 12 weeks, adding up to a total of 3 h. We did not train the teachers in the specific topics regarding the tools, we only provided them with basic information on how the website works and their task was to verify the entry to the page. This can cause that implementation to vary across schools, but there is not a clear way to measure the difference between them.

To implement the Game treatment, we personally went to each school and class to play the game with the students. We paid four one-academic hour visits (45 min), which added up to 3 h in total.

The follow-up survey was implemented in November 2018, after the program ended. This survey contained the same baseline survey questions, which allowed us to compare the effects of the program.

Table 4
Baseline summary statistics by treatment.

	(1) All Mean	SD	(2) Control Mean	SD	(3) Game Mean	SD	(4) Guide Mean	SD	(5) Web Mean	SD	(6) (2)–(3) diff	(7) (2)–(4) diff	(8) (2)–(5) diff	(9) (3)–(4) diff	(10) (3)–(5) diff	(11) (4)–(5) diff
Panel A. Demographics																
Participant's Age, years	11.08	3.10	11.23	3.10	11.18	2.99	11.17	3.25	10.72	3.04	0.05	0.05	0.05	0.05	0.05	0.05
Participant's Gender = 1, Female %	48.39	49.99	51.40	50.06	45.51	49.87	48.43	50.05	48.44	50.05	5.89	5.89	5.89	5.89	5.89	5.89
Father's Age, years	41.80	7.80	41.91	7.54	42.38	7.56	41.32	8.47	41.63	7.53	−0.48	−0.48	−0.48	−0.48	−0.48	−0.48
Father's Education, High School	0.34	0.48	0.34	0.47	0.35	0.48	0.38	0.49	0.31	0.46	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01
Father's Education, Technical	0.19	0.39	0.18	0.39	0.17	0.37	0.22	0.41	0.19	0.40	0.02	0.02	0.02	0.02	0.02	0.02
Father's Education, Professional	0.38	0.49	0.40	0.49	0.40	0.49	0.34	0.47	0.40	0.49	0.00	0.00	0.00	0.00	0.00	0.00
Father's Savings Account = 1, Yes	68.98	46.28	69.31	46.20	68.92	46.36	67.97	46.73	69.76	46.01	0.39	0.39	0.39	0.39	0.39	0.39
Stratum, (1–6)	2.61	0.96	2.61	0.98	2.62	0.98	2.60	0.94	2.62	0.94	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01
Household Size, n	4.36	1.74	4.27	1.74	4.34	1.77	4.44	1.82	4.36	1.62	−0.07	−0.07	−0.07	−0.07	−0.07	−0.07
Observations	1337		321		345		351		320		666	666	666	666	666	666

1. Stratum refers to a classification in strata of residential buildings that receive public utilities. Those with greater economic capacity pay more for public utilities and contribute so that the lower strata can pay their bills. The lowest stratum is 1 and the highest is 6.
2. Household size refers to the number of people who live in the child's household. This works as an approximation to the socioeconomic level.

Table 5
Timeline.

Month	March	July–October												November	Total Treatment Intensity
Week		1	2	3	4	5	6	7	8	9	10	11	12		
Guide	Baseline	20[a]	20	20	20	20	20	20	20	20	20			Follow-up	3.3 h
Website	Baseline	15	15	15	15	15	15	15	15	15	15	15	15	Follow-up	3 h
Game	Baseline			45			45			45			45	Follow-up	3 h
Control	Baseline													Follow-up	0 h

[a]Measure in minutes.

Table 6
Content and number of questions per grade.

Question subject	2nd Grade Age: 5–7	4th Grade Age: 8–11	7th Grade Age: 12–14	10th Grade Age: 15–18
Money and transactions	4	3	6	7
Planning and managing finances	3	3	3	5
Risk and reward			1	3
Financial landscape				2
Needs vs. Wishes	2	2		
Budget		1	1	1
Saving	1	2	1	1
Credit				1
Money sources	2	2		
Total	12	13	12	20

3.5. Data and variables

The variables collected in both in the baseline and the follow-up and that were used in this impact evaluation are the following:

3.5.1. Survey questions

Table 6 shows the number of questions per grade that measured financial knowledge and proficiency, as well as the main subjects they covered.

The questions were tailored to the program's material, objectives, and age[14] and were administered in all treatments and control groups. Some of the contents included in the survey questions are the following:

- Understand the concept of saving and be able to differentiate it from spending
- Understand the difference between things with monetary value and those without monetary value
- Understand the big difference between the things we want and the things we need
- Learn to compare prices to make better decisions
- Learn to create a monthly budget
- Learn to set financial goals

3.5.2. Raw knowledge score

Students' financial knowledge was calculated and scored on a scale of 0 to the maximum of possible answers for each grade, for example: a student in 2nd Grade could achieve a raw knowledge score from 0 to 12. We obtained one raw score at the baseline and one at the follow-up survey.

3.5.3. Knowledge score

Since all of the grades did not have the same number of questions, we decided to build a score on a scale of 0.0–5.0. The student financial knowledge could be scored on a scale of 0 to 100, however, we chose the 0.0–5.0 scale because this is the grading system in Colombia. Nonetheless, our results remained the same with both systems. We obtained one knowledge score at the baseline (pre-knowledge) and one at the follow-up survey (post-knowledge).

3.5.4. Knowledge delta score

The pre- and post-knowledge scores made it possible to identify skills and facts about financial education topics for each age

[14] Appendix C shows the content for each age/grade.

group in baseline and follow-up surveys, which helped us define our dependent variable.

We created the knowledge delta by using the two measurements: subtracting the post measurement from the pre measurement, in order to see the actual change or improvement in the knowledge score, which could go up to a maximum of 5.0. This variable allows for negative values, as the knowledge score may decrease.[15]

3.6. Independent variables

The key independent variables for this analysis are the treatment variable and a variety of demographic and sociodemographic characteristics.

We analyze the differences between those treated and those untreated, as well as the type of tool implemented (Game, Website, Guide). Our study also includes data on the type of school (public or private). In Colombia, the quality of public schools is not as high as that of private schools, which is why we believe that some of the differences, in terms of the children's results, depend on the type of school they attend (Castro Aristizabal et al., 2016).

We include sociodemographic characteristics related to the children, their parents, and their household as follows: age, gender (male, female), father's education (primary, high School, technical, professional), household size (number of people).

3.7. Behavioral predictions

The main purpose of this study is to analyze whether any of the tools actually work to increase the students' knowledge of personal finance. We expect for children exposed to any of the treatments to improve their skills more than those in the control group.

Hypothesis 1*: Treated participants have a higher knowledge delta than those in the control group.*

The additional purpose of this study is to determine which tool works the best from an educational perspective. Taking into account that there are three different tools designed for the same purpose, we expect the results for each of them to be different. We expect the knowledge delta to be higher for the more interactive tools (Website and Game). For the more traditional educational approach (Guide), we still expect an improvement, yet less so than with the other two.

Hypothesis 2*: A higher knowledge delta is expected if participants are treated. The predicted ordering of the knowledge delta across treatment is: Website>Game>Guide>Control*

Besides the direct program effects, we expect heterogeneous effects on type of school and grade. Given Colombia's education system, private school students' performance tends to be better than that of public school students. Factors such as part-time schooling, class size, behavioral standards and, in general, the level of school's efficiency, affect the children's education, meaning that both learning and improvement would also be more significant in children at private schools (Castro Aristizabal et al., 2016).

Regarding the grade, we expect that being treated nor the specific tool work exactly the same for every student since the age ranges from 6 to 18.

Hypothesis 3*: Participants from private schools improve their knowledge to a greater extent, than those in public schools. In addition, the effect of the treatment and the specific tool varies across grades.*

[15] The results are robust using the alternative raw score difference.

4. Results

4.1. Descriptive results

A total of 1337 students participated in both, the baseline and the follow-up survey. Participants' information corresponds to their knowledge and beliefs, as reflected in the questionnaire. Panel A of Tables 7 and 8 provides summary statistics of the variables obtained for each school and by grade. Female participation in the sample is of approximately 48%, and students' ages ranges from 7 for 2nd Grade, and 15 for 10th Grade, on average.

Data shows that the average value of some variables clearly depends on the students' socioeconomic level. The stratum[16] (socioeconomic) variable reveals differences between schools. On average, students from Nueva Colombia and Los Sauces belong to socioeconomic level 2, the GMMMC is closer to level 3, while students at the Anglo Americano school tend to belong to a slightly higher level 4. On average, each household has 4.36 persons, and this tends to be lower for private schools and higher for public schools. Regarding the average age of the father, the Nueva Colombia School (public and of a lower socioeconomic level) presents a lower value, while the Anglo Americano school (private and higher socioeconomic level) has the highest value. The fathers' education variable reveals that for Nueva Colombia, most fathers are high school graduates, and for the Anglo Americano school, most of them are professionals. In terms of the father's savings account, on average 36% of the parents of the Nueva Colombia School have a savings account, while 99% of the parents of the Anglo Americano school have one.

Panel B shows the outcome variables (Table 7, Table 8), beginning with the Pre-knowledge Score which refers to the initial score (0–5) obtained by the participant. The table shows that, on average, participants obtain a 3.09 score for the first measurement. At first glance, the evidence suggests that the performance of private schools is better in this respect. The lowest average score for this variable corresponds to 4th Grade and the Nueva Colombia School. The difference between private and public schools can be explained by a possible gap in the quality of education in Colombia, the economic resources of the school, its infrastructure, part time schooling, class size, behavior standards and the efficiency of the schools, in general.

The Post-knowledge Score shows the score obtained from the final questionnaire after the intervention, which ranges from 0.0 to 5.0. The table shows that the average score for the whole sample is 3.41 out of five points, which is higher than the score obtained for the pretest, and it maintains the same distribution across schools as the initial score. This shows an initial improvement of the knowledge score of 0.32 points on average for all the schools.

The Knowledge Delta Score shows the difference between the second and the first score, with the measurement ranging from −3.04 to 4.62 for the full sample. We can see that the lowest delta measurement of 0.06 is for the Nueva Colombia School, which is the only public school. On the other hand, the highest delta is 0.49 points, for GMMMC School, which is a private one.

4.2. Empirical results

Fig. 1 shows, the average knowledge by school, before and after the treatment was implemented. The average score for all schools is 3.1, of a maximum of 5.0 before the treatment, and

[16] Socio-economic stratification is a classification in strata of residential buildings that must receive public utilities. Those with more economic capacity pay more for public utilities and contribute so that the lower strata can pay their bills. The lowest stratum is 1 and the highest is 6 (DANE).

Table 7
Descriptive statistics by school.

	(1) Nueva Colombia Mean	(2) Los Sauces Mean	(3) GMMMC Mean	(4) Anglo Americano Mean	(5) All Mean
Panel A. Control Variables					
Participant's Age, years	11.09	11.36	10.64	11.12	11.08
Participant's Gender = 1, Female %	47.60	44.96	57.69	43.94	48.39
Father's Age, years	39.05	40.83	41.97	46.23	41.80
Father's Education, High School	0.59	0.44	0.26	0.00	0.34
Father's Education, Technical	0.16	0.25	0.28	0.04	0.19
Father's Education, Professional	0.08	0.20	0.44	0.95	0.38
Father's Savings Account = 1, Yes	36.33	66.84	77.01	99.60	68.98
Stratum, (1–6)	2.03	2.04	2.65	4.16	2.61
Household Size, n	4.80	4.35	4.40	3.85	4.36
Panel B. Outcome Variables					
Pre-knowledge (0–5)	2.76	3.16	2.95	3.55	3.09
Post-knowledge (0–5)	2.82	3.52	3.43	3.95	3.41
Knowledge Delta	0.06	0.36	0.49	0.40	0.32
Observations	334	427	312	264	1337

1. Stratum refers to a classification in strata of residential buildings that receive public utilities. Those with greater economic capacity pay more for public utilities and contribute so that the lower strata can pay their bills. The lowest stratum is 1 and the highest is 6.
2. Household size refers to the number of people who live in the child's household. This works as an approximation to the socioeconomic level.
3. Pre-knowledge refers to the score obtained for the pretest (from 0 to 5 points).
4. Post-knowledge refers to the score obtained for the posttest (from 0 to 5 points).
5. Knowledge Delta refers to the difference between the pretest and the posttest score (from −5 to 5 points).

Table 8
Descriptive statistics by grade.

	(1) 2nd Grade Mean	(2) 4th Grade Mean	(3) 7th Grade Mean	(4) 10th Grade Mean	(5) All Mean
Panel A. Control Variables					
Participant's Age, years	6.96	9.09	12.18	15.19	11.08
Participant's Gender = 1, Female %	47.65	49.71	50.94	44.87	48.39
Father's Age, years	38.67	39.65	42.94	45.33	41.80
Father's Education, High School	0.27	0.38	0.37	0.35	0.34
Father's Education, Technical	0.20	0.18	0.19	0.20	0.19
Father's Education, Professional	0.48	0.38	0.35	0.35	0.38
Father's Savings Account = 1, Yes	75.40	68.51	68.37	64.75	68.98
Stratum, (1–6)	2.72	2.61	2.57	2.57	2.61
Household Size, n	4.25	4.47	4.33	4.36	4.36
Panel B. Outcome Variables					
Pre-knowledge (0–5)	2.99	2.93	3.17	3.25	3.09
Post-knowledge (0–5)	3.58	3.40	3.44	3.25	3.41
Knowledge Delta	0.59	0.47	0.27	0.00	0.32
Observations	277	348	371	341	1337

1. Stratum refers to a classification in strata of residential buildings that receive public utilities. Those with greater economic capacity pay more for public utilities and contribute so that the lower strata can pay their bills. The lowest stratum is 1 and the highest is 6.
2. Household size refers to the number of people who live in the child's household. This works as an approximation to the socioeconomic level.
3. Pre-knowledge refers to the score obtained for the pretest (from 0 to 5 points).
4. Post-knowledge refers to the score obtained for the posttest (from 0 to 5 points).
5. Knowledge Delta refers to the difference between the pretest and the posttest score (from −5 to 5 points).

3.4 following the treatment. There is a significant difference in means of 0.3 points in the knowledge score. However, the figure shows that only the last three schools (private) have a significant difference in improvement. On the other hand, the public school (N Colombia) shows no apparent improvement between the pre- and post-measurement. At first glance, this coincides with our *Hypothesis 3*: children in private schools show greater improvement compared to those in public ones. However, we cannot yet say if said improvement is due to the treatment or because of the pre-existing differences between private and public schools. This may be the result of a quality gap in Colombia's education system, its infrastructure and teaching staff, which contrasts with the higher quality of private schools and generates a good environment for programs to create impact, but this should be analyzed more thoroughly.

To further analyze the treatment effect and the school type effect, we estimated model (1). Using a linear regression model (OLS), we sought to identify the effect of being treated on the knowledge delta and the determinants of the knowledge level, taking into account variables such as being treated, the school type and the interaction term between treatment and private school. X_i corresponds to individual characteristics (student's age and gender, the father's education, and the household size) of student i and is included to increase precision. $\epsilon_{ijs}js$ is a clustered error term for student i, classroom level j and school s. Clustering

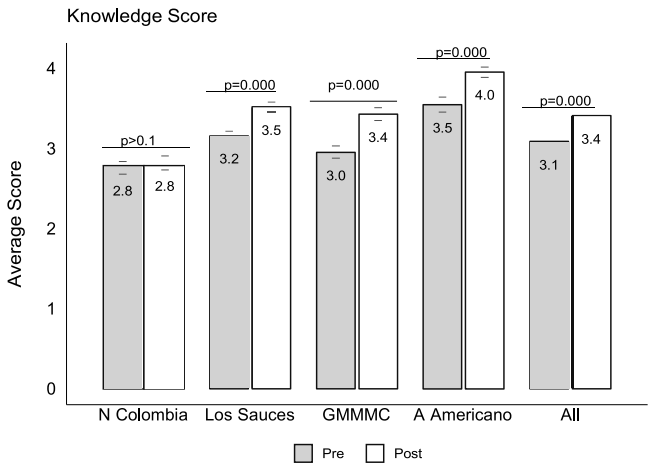


Fig. 1. Knowledge delta by school.

Table 9
Effect of financial education program on knowledge delta — type of school.

Variables	(1) All	(2) Public	(3) Private
Treated = 1, Yes	0.040	0.124	0.019
	(0.224)	(0.182)	(0.088)
School Type = 1, Private	0.380*		
	(0.218)		
Treated x School Type = 1, Private	−0.030		
	(0.244)		
Participant's Age, years	−0.018	0.089	−0.064
	(0.051)	(0.097)	(0.049)
Participant's Gender = 1, Female	−0.035	−0.159	−0.022
	(0.057)	(0.161)	(0.062)
Father's Education = 1, High School	−0.238**	−0.573***	0.041
	(0.113)	(0.192)	(0.089)
Father's Education = 2, Technical	−0.170	−0.417*	0.080
	(0.120)	(0.229)	(0.121)
Father's Education = 3, Professional	−0.158	0.136	0.058
	(0.114)	(0.297)	(0.097)
Household Size, n	0.013	0.016	0.014
	(0.023)	(0.037)	(0.027)
Constant	0.464	−0.371	0.975**
	(0.453)	(0.874)	(0.403)
Observations	1,136	257	879
R-squared	0.074	0.144	0.052
Grade FE	Yes	Yes	Yes
H_0 pvalue: Control = Treated	0.857	0.505	0.832
H_0 pvalue: Public = Private	0.0859		
H_0 pvalue: Treated = TreatedxPrivate	0.879		
H_0 pvalue: Private = TreatedxPrivate	0.356		

1. Clustered standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.
2. Treated refers to the assignment to any of the treatment tools.
3. Household size refers to the number of people living at the child's house.

is done at classroom level as the program's implementation.

$$\Delta Knowledge_i = \beta_0 + \beta_1 treated_i + \beta_2 school_type_i + \beta_3 treatedxschool_type_i + \beta_4' X_i + \epsilon_{ijs} \quad (1)$$

Column 1 of Table 9 shows the results of model (1) for the whole sample. The estimation shows that there is no effect for being treated. Even though the direction is positive (being treated would increase the delta score), this result is not significant. At first glance, this would lead us to think that the program was not effective in terms of improving personal finance knowledge in students, thus rejecting Hypothesis 1. There might be mixed effects in this lack of effect of the treatment. First, not all the tools are effective and can cancel out the effect. Second, the implementation of all the tools is different, causing heterogeneous effectiveness. Finally, there might also be mixed effects by school or grade.

On the other hand, being in a private school increases the knowledge score delta by 0.380 decimals, showing that there is a pre-existing differential effect among students in private schools against public schools. This result coincides with the findings of Lusardi et al. (2010), where the better quality of the institution, measured according to the teachers' interest in students, improves the probability of providing correct answers. The interaction effect is not significant either.

The only significant demographic characteristics that cause a decreasing impact in the knowledge delta is the father's education, when it is no higher than high school. When we divide the sample by public (Column 2) and private (Column 3), we see that the father's education is relevant only for the public school, which is an expected result. The treatment lacks significant effect when the sample is divided.

Table 10 shows the analysis of the effect of being treated by grades, using model (1). Column 1 shows that, in 2nd Grade, being treated has a detrimental impact on the knowledge delta. This effect might be driven by the effect of the different tools altogether. The only relevant demographic variable that decreases the knowledge improvement is the household size.

Column 2 shows that 4th Grade does not show a significant impact on the knowledge delta for those who are being treated, and there are no differential effects in terms of type of school. Household size continues to be a decreasing and relevant factor of knowledge delta. In neither 2nd Grade nor 4th did we find that the type of school alone is relevant, suggesting that younger kids do not suffer as much from the differences in the type of school, because they are just at the early stages of their education.

Column 3 corresponds to 7th Grade and shows that the treatment makes no difference. However, the school type is relevant as a pre-existing condition; being in a private school increases the knowledge delta by 0.544. In addition to this, we can see that comparing $\beta_2 = \beta_3$ is significant, therefore, there are heterogeneous effects from being treated, but not in the way we expected.

Column 4 shows 10th Grade results, this is the only grade in which the treatment seems to increase the knowledge delta in 0.769. However, the school type is relevant as a pre-existing condition; being in a private school increases the knowledge delta by 1.209. On the other hand, we see an heterogeneous and significant impact of the treatment. This suggests that being treated has a positive impact, but this result is driven by the improvement of public schools students, since receiving treatment in private schools actually reduces the improvement (−0.771).

Results for 7th and 10th suggest that, after several school years, the quality or type of the school reveals an intrinsic pre-existing difference on children's knowledge; however, being involved in a program can improve the results in public schools.

As a result regarding *Hypothesis 1*, we can see that being treated had no significant overall effect on the students, which we believe is due to the fact that not all the tools worked efficiently or at least not in every grade, or that they were not correctly implemented.

As for *Hypothesis 3*, we see that the impact of the treatment indeed reveals differences, depending on the grade and the type of school. We found a positive improvement in treated public school children, specifically in 10th grade.

$$\Delta Knowledge_i = \theta_0 + \theta_1 tool_i + \theta_2 school_type_i$$

Table 10
Effect of financial education program on knowledge delta — Grade.

VARIABLES	(1) 2nd Grade	(2) 4nd Grade	(3) 7nd Grade	(4) 10nd Grade
Treated = 1, Yes	−0.862*	0.121	0.087	0.769***
	(0.473)	(0.150)	(0.149)	(0.072)
School Type = 1, Private	−0.092	−0.220	0.544***	1.209***
	(0.171)	(0.136)	(0.117)	(0.077)
Treated x School Type = 1, Private	0.791	0.205	−0.249	−0.771***
	(0.508)	(0.217)	(0.191)	(0.132)
Participant's Age, years	−0.071	−0.038	0.070	−0.036
	(0.134)	(0.073)	(0.092)	(0.091)
Participant's Gender = 1, Female	0.125	−0.046	−0.158	−0.084
	(0.136)	(0.135)	(0.120)	(0.103)
Father's Education = 1, High School	0.205	−0.196	−0.483*	−0.205
	(0.213)	(0.151)	(0.251)	(0.151)
Father's Education = 2, Technical	0.246	−0.054	−0.387	−0.100
	(0.255)	(0.272)	(0.246)	(0.110)
Father's Education = 3, Professional	0.195	−0.007	−0.203	−0.246*
	(0.175)	(0.244)	(0.222)	(0.127)
Household Size, n	−0.040*	0.063*	−0.009	−0.002
	(0.020)	(0.032)	(0.046)	(0.050)
Constant	1.191	0.552	−0.486	−0.325
	(0.952)	(0.646)	(1.227)	(1.329)
Observations	238	291	325	282
R-squared	0.104	0.053	0.062	0.107
H_0 pvalue: Control = Treated	0.0883	0.436	0.569	3.96e−08
H_0 pvalue: Public = Private	0.599	0.127	0.000363	2.83e−10
H_0 pvalue: Treated = TreatedxPrivate	0.106	0.813	0.312	1.44e−06
H_0 pvalue: Private = TreatedxPrivate	0.138	0.160	0.00952	4.52e−08

1. Clustered standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.
2. Treated refers to the assignment to any of the treatment tools.
3. Household size refers to the number of people living at the child's house.

$$+ \theta_4 grade_i + \theta_3' X_i + \epsilon_{ijs} \quad (2)$$

To analyze each tool in particular, and their effects on the knowledge delta, we estimated model (2). Using a linear regression model (OLS), we sought to identify the effect of each tool on the knowledge delta and the determinants of the improvement in knowledge. X_i corresponds to individual characteristics (student's age and gender, the father's education, and the household size) of student i and is included to increase precision. ϵ_{ijs}*js* is a clustered error term for student i, classroom level j and school s. Clustering is done at classroom level as the program's implementation.

Table 11 Column 1 shows the ordinary least squares estimation for the whole sample. As before, the estimation shows that there is a significant school type effect for the full sample. We observe that none of the tools appear to be effective; however, this might change when dividing the sample by grade and type of school.

Column 2 shows the results of the model for 2nd Grade and Public School, where the Game and the Guide, instead of improving knowledge, deteriorate it when compared to not receiving treatment. However, the order of the tools supports our Hypothesis 2, where the best learning tool is the Website. On the other hand, Column 3 shows results for 2nd Grade Private School, where none of the tools yield significant improvement. These results show a clear differential effect between types of schools, for the 2nd Grade.

Column 4 shows that, for 4th Grade-Public, the Web is significant and increases the knowledge delta by 0.397 points, compared to the control group, but is also better than the Guide. The Website treatment also increases knowledge in 0.416, compared to the control group, and is actually better than any of the other tools. In contrast, for the private school (Column 5) the Game improves financial knowledge. This supports our *Hypothesis 2*, at least for 4th Grade, where the interactive tools have a positive effect on the outcome; however, the tools that are effective are different by type of school.

Column 6 shows that, for 7th Grade-Public, the Guide increases the outcome variable while, for the Private School, the Guide actually decreases knowledge (Column 7), showing a clear distinction between the types of schools and the way students learn in each of them. These results might explain the former lack of overall treatment effect on the 7th Grade, given that the Guide's effect cancels it out.

Finally, Column 8 and 9 show the results for 10th Grade, in both public and private schools. The differences between these types of schools are evident because all of the tools improved financial knowledge in the public school, while none of them were relevant for the private schools. The higher grades of the public schools (7th and 10th) show that the more traditional tool, the Guide, had a stronger effect, which contradicts our initial *Hypothesis 2*. Since the Guides were designed to be completed at home, the time and quality of the implementation could vary to a great extent across households. The effect that this has, especially on younger students that require more assistance, may explain this matter.

In conclusion, these results do not fully support the direction we expected for *Hypothesis 2* to take, these results support *Hypothesis 3* because we find several heterogeneous effects among grades, and more importantly, among the types of schools.

5. Conclusion

This paper explores the effect of a randomized financial education program implemented in Colombia, and consisting of three learning tools (a Game, a Guide and a Website), on students' financial knowledge. We present a comparison of pretest and

Table 11
Effect of financial education program tools on knowledge delta — type of school and grade.

Variables	(1) All	(2) 2nd Grade Public	(3) 2nd Grade Private	(4) 4nd Grade Public	(5) 4nd Grade Private	(6) 7nd Grade Public	(7) 7nd Grade Private	(8) 10nd Grade Public	(9) 10nd Grade Private
Tool = 1, Game	0.138	−1.047***	0.058	0.250	0.443**	−0.088	0.150	0.730**	0.157
	(0.105)	(0.094)	(0.244)	(0.120)	(0.168)	(0.066)	(0.154)	(0.162)	(0.093)
Tool = 2, Guide	−0.127	−1.834***	−0.158	−0.134	0.136	0.742**	−0.410***	0.913***	−0.109
	(0.109)	(0.133)	(0.185)	(0.160)	(0.192)	(0.182)	(0.123)	(0.065)	(0.091)
Tool = 3, Web	0.047	0.052	0.006	0.416**	0.249	0.175	−0.208	1.191***	0.007
	(0.108)	(0.086)	(0.265)	(0.126)	(0.170)	(0.199)	(0.136)	(0.189)	(0.069)
School Type = 1, Private	0.350***								
	(0.120)								
Participant's Age, years	−0.014	0.056	−0.221*	0.085	−0.063	−0.031	0.145	0.236	−0.109
	(0.051)	(0.289)	(0.113)	(0.144)	(0.104)	(0.228)	(0.081)	(0.142)	(0.090)
Participant's Gender = 1, Female	−0.031	−0.006	0.084	−0.068	−0.030	−0.660*	−0.039	−0.312	−0.046
	(0.059)	(0.575)	(0.116)	(0.161)	(0.200)	(0.267)	(0.116)	(0.169)	(0.124)
Father's Education = 1, High School	−0.223*	0.737	0.106	−0.272*	0.248	−1.293	−0.003	−0.776**	0.009
	(0.113)	(0.447)	(0.292)	(0.090)	(0.445)	(0.577)	(0.124)	(0.147)	(0.105)
Father's Education = 2, Technical	−0.148	0.739	0.153	−0.099	0.301	−1.130	0.059	−0.395	0.032
	(0.120)	(0.631)	(0.282)	(0.293)	(0.625)	(0.643)	(0.196)	(0.473)	(0.112)
Father's Education = 3, Professional	−0.153	0.847	0.036	0.649	0.333	−0.170	0.136	−0.297	−0.096
	(0.110)	(0.509)	(0.205)	(0.331)	(0.507)	(0.619)	(0.143)	(0.486)	(0.097)
Household Size, n	0.014	−0.052	−0.067**	0.054	0.071	−0.042	0.002	0.015	0.001
	(0.023)	(0.029)	(0.029)	(0.062)	(0.041)	(0.042)	(0.056)	(0.127)	(0.051)
Constant	0.450	−0.012	2.381**	−0.516	0.176	1.625	−1.336	−4.247	1.795
	(0.419)	(1.534)	(0.778)	(1.044)	(1.336)	(2.829)	(0.991)	(1.858)	(1.305)
Observations	1,136	46	192	83	208	76	249	52	230
R-squared	0.085	0.360	0.039	0.173	0.067	0.271	0.067	0.258	0.039
Grade FE	Yes	No	No	No	No	No	No	No	No
H_0 pvalue: Control = Game	0.194	0.00155	0.818	0.129	0.0249	0.272	0.352	0.0204	0.123
H_0 pvalue: Control = Guide	0.249	0.000822	0.411	0.465	0.495	0.0267	0.00761	0.000773	0.257
H_0 pvalue: Control = Web	0.663	0.589	0.981	0.0455	0.174	0.443	0.156	0.00803	0.919
H_0 pvalue: Game = Guide	0.0150	0.00129	0.385	0.00761	0.248	0.00788	0.00547	0.222	0.0490
H_0 pvalue: Game = Web	0.381	0.00103	0.870	0.00577	0.275	0.145	0.0526	0.0310	0.231
H_0 pvalue: Guide = Web	0.115	0.000142	0.549	0.000839	0.677	0.00670	0.232	0.190	0.130

1. Clustered standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.
2. Tool refers to the assigned treatment.
3. Household size refers to the number of people living at the child's house.

posttest scores for a sample of 1337 students from four Colombian schools, for four academic levels. We obtain one of the few measurements that currently exist, for the level of financial literacy in Colombia, and provide information about one of the few programs implemented in this field.

We found that the type of school (public or private) is decisive in terms of the improvement of students' knowledge, and is still today a pre-existing condition that makes a difference in terms of children's performance. In line with Mandell (2008), we did not find a clear increase in financial literacy scores after implementing a personal finance program. In particular, we did not find a significant treatment effect upon comparison with the control group; for public schools, we found a significant positive effect on the improvement of financial knowledge in those who received treatment. Regarding the tools included in the financial education program, the results are heterogeneous by type of school (public/private) and by academic degree. It is not possible to draw general conclusions about the effectiveness of the tools because it depends on those two variables, however, this is not a bad outcome; it actually helps to understand the importance of an adjusted program for different types of students, which goes in line with what previous literature has found about the different effects depending on the target group (Kaiser and Menkhoff, 2017). Our results, however, contradicts those of Iterbeke et al. (2020): we do find that differentiated instruction during the learning process can be effective, even if significant pre-existing differences still prevail.

On the one hand, work should be conducted to improve the quality of general education in public schools, where it is normal for schools to have students on a 'double shift' basis, some in the morning and others in the afternoon, in order to cover more students. However, this affects the quality of education because contact time with the training processes is shorter than in private schools, where students have full school days. In addition, class size tends to be larger in public schools, the behavioral standards are different, and public schools generally tend to be less efficient. The study shows that the performance of children in private schools without treatment is, on average, better than in the public school, revealing a pre-existing condition that works against the latter. Nonetheless, this study shows that creating and implementing an educational program at public schools can have a positive impact to reduce the existing differences. In our sample, particularly the higher grades of the public schools were able to benefit from the treatments, especially the 10th grade, which attained positive improvement by using all the learning tools. For students at public schools, having a high quality program can have a bigger impact since their family conditions and general environment outside school provide less opportunities for improvement.

Based on the data collected on levels of financial literacy and improvement, we can see that grade is a relevant factor in determining an educational program. It is indeed useful to intervene at a school level as children are at an ideal age and within a learning environment. However, the program should be differentiated, because motivation and learning processes vary across ages (Amagir

Table A.12
Attrition by school.

	(1)		(2)		(3)
	5 Schools		4 Schools		Difference
	Mean	Std. Dev.	Mean	Std. Dev.	(2)-(1)
Control, %	23.81	42.60	24.02	42.73	0.21
Game, %	25.57	43.63	25.57	43.64	0.00
Guide, %	23.85	42.62	23.92	42.67	0.07
Web, %	26.77	44.29	26.50	44.14	−0.27
2nd Grade, %	24.01	42.72	23.76	42.57	−0.25
4th Grade, %	24.21	42.84	24.59	43.07	0.38
7th Grade, %	26.13	43.94	26.08	43.92	−0.05
10th Grade, %	25.65	43.68	25.57	43.64	−0.08
School Type = 1, Public %	44.90	49.75	28.87	45.33	−16.03***
School Type = 1, Private %	55.10	49.75	71.13	45.33	16.03***
Observations	2499		1936		2499

Table A.13
Attrition after consent.

	(1)		(2)		(3)
	All		Consent		Difference
	Mean	Std. Dev.	Mean	Std. Dev.	(2)-(1)
Control, %	23.89	42.65	23.19	42.22	−0.7
Game, %	25.48	43.59	25.84	43.79	0.36
Guide, %	24.10	42.78	26.44	44.12	2.34
Web, %	26.54	44.17	24.52	43.03	−2.02
2nd Grade, %	24.04	42.75	20.41	40.32	−3.63**
4th Grade, %	25.05	43.34	25.31	43.50	2.26
7th Grade, %	25.48	43.59	27.57	44.70	2.09
10th Grade, %	25.42	43.56	26.71	44.26	1.29
School Type = 1, Public %	28.61	45.21	27.57	44.70	−1.04
School Type = 1, Private %	71.39	45.21	72.43	44.70	1.04
Observations	1884		1509		1884

Table B.14
Full process timeline.

Year	Semester	Activity
2011	1	Review of surveys and programs to define age groups and content
	2	
2012	1	
	2	
2013	1	Pilot test survey in schools and adjustment of the questions. New questions are added
	2	Development of the Guides
2014	1	
	2	Review of the Guides by the Ministry of Education
2015	1	Guides adjustment
	2	
2016	1	Guides pilot test (Colegio Colsubsidio)
		Guides adjustment
	2	Website development
2017	1	DevelopmentL of the Website and Development of the Game
	2	Pilot Test Website and Game + Settings Website and Game
2018	1	School Sample Selection Baseline survey
	2	Implementation at Schools Follow-up Survey
2019	1	Database construction and validation
	2	Modeling and Results
2020	1	
	2	Research and Writing

et al., 2018; Kaiser and Menkhoff, 2020). Moreover, students at higher grades can also benefit from a financial education program since they have more clear sense of money or the importance of personal finance, compared to younger kids.

Finally, financial literacy is strongly related to sociodemographic characteristics as well as family financial sophistication (Lusardi et al., 2010). We also found that the fathers' level of education is an influencing factor of knowledge improvement. Thus, as stated by Van Campenhout (2015), the parents' role in

Table B.15
Surveys.

Surveys	Country	Year	References
PISA 2012 Financial Literacy Framework	OECD	2012	Programme for International Student Assessment. (2011). OECD — Programme for International Student Assessment. http://www.pisa.oecd.org/dataoecd/8/43/46962580.pdf
Financial capability in the United States – National Survey	USA	2009	Financial Literacy and Education Commission. (2006). MyMoney.gov. http://www.mymoney.gov/sites/default/files/downloads/ownership.pdf
Etude sur l'argent et les problématiques financiéres auprés des jeunes 15–20 ans	France	2006	La Finance pour tous. (November of 2006). La Finance pour tous. http://www.lafinancepourtous.com/Une-enquete-sur-les-jeunes-et-l.html
Campaign for Economic Literacy	USA	2005	Council for Economic Education. (2005). Council for Economic Education. http://www.councilforeconed.org/cel/results.php#interest
Estudio de Educación Financiera en niños de escuelas primarias públicas	Mexico	2008	Secretaría de Educación Pública de México. (2008). Secretaría de Educación Pública. http://www.sep.gob.mx/work/models/sep1/Resource/93115/cuestionario.swf
Financial Capability in the UK: establishing a baseline	UK	2006	Financial Services Authority. (2006). FSA. http://www.fsa.gov.uk/pubs/other/fincap_baseline.pdf
Jump$tart Coalition Survey of Personal Financial Literacy Among Students	USA	2008	JumpStart.org. (2006). SavingAdvice.com. http://www.savingadvice.com/tools/quizzes/jumpstart-financial-literacy.html
Encuesta de carga y educación financiera de los hogares (Banco República)	Colombia	2010	Ministerio de Hacienda, Ministerio de Educación, Banco de la República, Superintendencia Financiera, Fondo de Garantías de Instituciones Financieras, Fondo de Garantías de Entidades Cooperativas, et al. (2010). Estrategia Nacional de Educación Económica y Financiera.
Encuesta de Educación Financiera 2010 -Primer Semestre	Mexico	2010	Programa Educatec Financieramente. (2010). https://docs.google.com/fileview?id=0B339DBZL7HvhZWVmNWM0NzQtMWRiZS00NzNhLWI2YzUtMzBiMzFlMTFkYjQ2&hl=en
Financial Literacy Quiz - FINRA Investor Education Foundation	USA	2010	FINRA. (2010). FINRA Investor Education Foundation. http://www.usfinancialcapability.org/index.html
Educational programs			
Educação Financeira nas Escolas	Brazil	2010	Comité para la regulación y supervisión de mercados financieros, capital, seguros, planes de pensiones y capitalización (COREMEC). (2010). Vida e Dinheiro. http://www.vidaedinheiro.gov.br/EducacaoFinanceira/Default.aspx
Técnicas financieras para toda la vida	USA	2010	Wells Fargo. (2010). Técnicas financieras para toda la vida. http://www.elfuturoentusmanos.org/es/recursos-para-la-instruccion.html
Education Enterprise	Australia	2009	Applied Research and Consulting LLC. (2009 de December of 2009). FINRA - Investor Education Foundation. http://www.finrafoundation.org/web/groups/foundation/@foundation/documents/foundation/p120536.pdf
Children's Financial Network – CFN	USA	2011	Children's Financial Network. (2011). Children's Financial Network Inc. http://www.childrensfinancialnetwork.com/index.cfm
NEFE High School Financial Planning Program	USA	2007	National Endowment for Financial Education. (2007). NEFE High School Financial Planning Program. http://hsfpp.nefe.org/channels.cfm?chid=97&tid=1&deptid=14
Citi Group Financial Education Curriculum	USA	2011	Citi Group. (2011). Citi - Citi never sleeps. http://www.citigroup.com/citi/financialeducation/curriculum/kids_sp.htm
Other sources			
Summary of financial education programmes in schools in selected OECD and non-OECD member countries	Australia Malaysia England Netherlands	2008	Mundy, S. (2008). International Gateway for Financial Education. http://www.financial-education.org/dataoecd/38/43/44509425.pdf
Framework for the development of financial literacy baseline surveys	OECD	2009	Kempson, E. (2009). Framework for the Development of Financial Literacy Baseline Surveys: A First International Comparative Analysis. Bristol: OECD Publishing.
HSBC Financial Literacy Programme: Global Impact Report	USA	2008	HSBC. (2008). SIFE. http://www.sife.org/aboutsife/News/Documents/FINALGlobalImpactReport.pdf
Rich Kid Smart Kid (RKSK)	USA	2010	The Rich Dad Company. (2010). Rich kid smart kid. http://richkidsmartkid.com/about.html

their children's education is indeed important, as they can help them achieve better results in the future.

Even though, there is no significant treatment effect, we did find that there are some tools that work better than others in improving scores, depending on the target group. The interactive tools, such as the Game and the Website, have a greater effect on some students, particularly on the younger ones; because of the way they are implemented, students become more engaged than with a traditional learning tool such as the Guide. This goes in line with what Amagir et al. (2018) states about an interactive approach through experiential learning being a promising method for financial education. For some other groups, especially older kids at public schools, the Guide appears to be the best learning tool. We can say that for population groups such as the ones analyzed, and before doing a generalized program, it is better to understand the heterogeneity among groups to design a specific and effective learning tool.

Although this is a sample of only four schools in Bogota, Colombia, which by no means pretends to be representative, it

Table B.16
Thematic contents.

Subject	PISA				Others
	Money and transactions	Planning and managing finances	Risk and reward	Financial landscape	
Financial capability in the United States – National Survey. (65 questions)	X	X	X	X	X
Etude sur l'argent et les problématiques financiéres auprés des jeunes 15–20 ans. (28 questions)	X	X			X
Campaign for Economic Literacy (20 questions)	X		X	X	
Estudio de Educación Financiera en niños de escuelas primarias públicas (10 questions)	X	X	X		
Financial Capability in the UK: establishing a baseline	X	X	X	X	X
Jump$tart Coalition Survey of Personal Financial Literacy Among Students (31 questions)	X	X	X	X	
Encuesta de carga y educación financiera de los hogares (10 questions)			X		
Encuesta de Educación Financiera 2010-Primer Semestre. (29 questions)	X		X	X	X
Financial Literacy Quiz (5 questions)	X	X	X		
Técnicas financieras para toda la vida – Guía Niños	X	X			
Técnicas financieras para toda la vida – Guía Adultos	X				
Técnicas financieras para toda la vida – Guía Adultos jóvenes	X	X			X
Técnicas financieras para toda la vida – Guía Adolescentes	X	X	X		
Education Enterprise	X	X			
NEFE High School Financial Planning Program.		X	X		X
Citi Group Financial Education Curriculum.	X	X	X		X
Framework for the development of financial literacy baseline surveys.	X	X		X	
Rich Kid Smart Kid	X	X	X		

Table B.17
Age selection.

Age (years)	OECD				Etude sur l'argent et les problema-tiques financiéres	Técnicas financieras para toda la vida	Citi Group	RKSK	PISA	Uniandes
	Australia	Malaysia	England	Netherlands						
5										A
6	A		A	A			A	A		A
7	A	B	A	E			A	A		A
8	A	B	B	E			B	B		B
9	B	B	B	B		B	B	B		B
10	B	B	B	B		B	B	B		B
11	C	B	B	C		C	C	C		B
12	C	B	D	C		C	C	C		C
13	D	D	D			C	C	C		C
14	D	D	D			D		D		C
15		D	D		D	D		D	D	D
16		D	D		D	D		D	D	D
17										D
18										D

Each of the letters in the table represents the division that the entities made for their financial education courses or surveys.

is a first example of what happens with Colombian children's financial education. From this first baseline and implementation, new challenges arise in terms of designing and implementing policies and programs that may improve the educational situation of children regarding their personal finances. Taking this step at an early age can ensure that they make better financial decisions in the future.

CRediT authorship contribution statement

Santiago Rodriguez-Raga: Term, Conceptualization, Methodology, Investigation, Writing – review & editing, Supervision. **Natalia Martinez-Camelo:** Methodology, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization.

Appendix A

See Tables A.12 and A.13.

Appendix B

See Tables B.14–B.17.

Table C.18
Thematic contents by age and grade.

Age: 5–7 Grade: 2	Age: 8–11 Grade: 4
1. Understand the value of money and recognize the denomination of coins 2. Understand the role of money in exchange 3. Understand that people work for money and know some types of work 4. Understand the concept of saving and be able to differentiate it from spending 5. Understand the difference between things with monetary value and those without monetary value 6. Understand the big difference between the things we want and the things we need	1. Recognize the denomination of banknotes 2. Start saving through allowance 3. Begin to understand how spending decisions are made 4. Learn to make a simple budget 5. Learn to compare prices to make better decisions 6. Understand the importance of banks
Age: 12–14 Grade: 7	**Age: 15–18 Grade: 10**
1. Recognize those goods and services that are essential for our wellbeing 2. Learn to set short-term financial goals 3. Start saving 4. Understand the concept of compound interest 5. Understand how the credit card works	1. Understand the use of the credit card 2. Learn the role and impact of taxes 3. Learn to create a monthly budget 4. Learn to set medium-term financial goals 5. Understand the difference between the employee and the entrepreneur 6. Understand the basic concepts of economics (law of supply and demand, unemployment rate, exchange rate, inflation, interest rates, GDP) 7. Understand the role of insurance and existing types of insurance 8. Know the types of existing investments

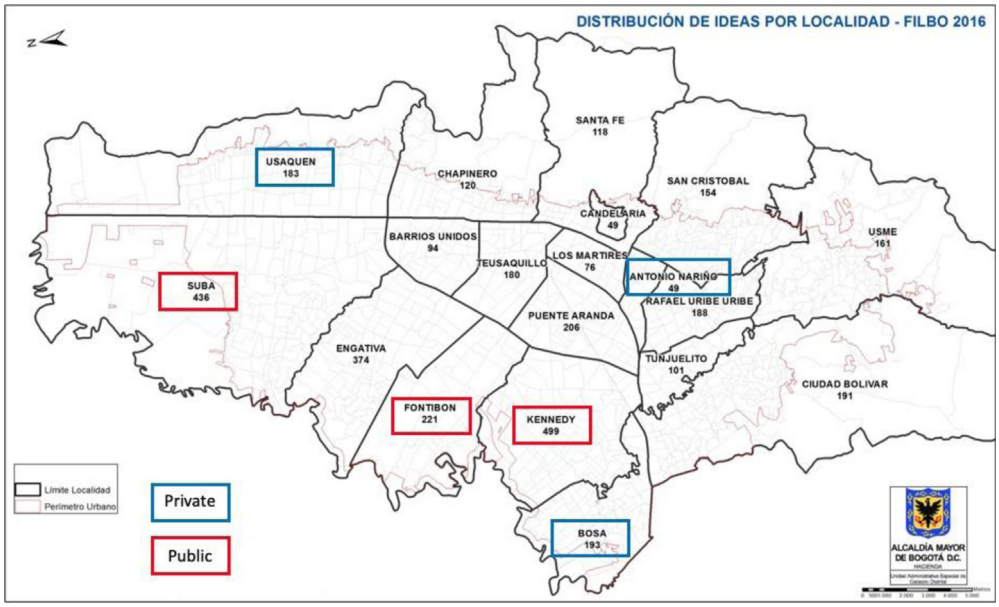


Fig. D.2. Localities in Bogota, Colombia.

Appendix C

See Table C.18.

Appendix D

See Fig. D.2.

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